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# Nowcasting Economic Activity in Afghanistan

Hany Abdel-Latif, Abdullah AlHassan, Karim Badr, Iyke Maduako

**WP/26/238**

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Middle East and Central Asia Department

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**Prepared by Hany Abdel-Latif, Abdullah AlHassan, Karim Badr, Iyke Maduako***

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**ABSTRACT:** Sound economic policy formulation requires high frequency economic activity data. This is particularly challenging in low-income, fragile, and conflict-affected states, such as Afghanistan, where quarterly GDP data is unavailable and annual GDP figures are published with considerable delays. This paper utilizes an innovative approach for estimating quarterly economic activity under these constraints, facilitating timely economic assessments in Afghanistan. Quarterly GDP data are interpolated from annual figures, utilizing money supply as a proxy indicator of economic activities due to Afghanistan's predominately cash-based economy. By leveraging machine learning, traditional (macroeconomic), non-traditional (satellite) data, the approach provides real-time insights of the underlying direction of the Afghan economy at relatively high frequency and with partial information.

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WORKING PAPERS

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Hany Abdel-Latif, Abdullah AlHassan, Karim Badr, Iyke Maduako

# Nowcasting Economic Activity in Afghanistan*

Hany Abdel-Latif[a], Abdullah AlHassan[a], Karim Badr[a], Iyke Maduako[a]

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## 1. Introduction

Limited and delayed data often hampers real-time monitoring of economic activity. This challenge is particularly acute in many low-income, fragile, and conflict-affected states, where GDP data is available only on an annual basis and is frequently subject to significant publication delays. These economies typically face limited institutional capacity, constrained resources, and widespread informality, making timely compilation and dissemination of macroeconomic statistics difficult. In such contexts, the absence of timely information can severely undermine the effectiveness of policy responses, especially during periods of crisis.

In Afghanistan, real-time data on economic activity is especially scarce. GDP figures are published only annually, with a 6–8 months lag and there is limited timely information on economic sectors, industrial production, or business activity at higher frequencies. Given that tracking economic activity is essential for monitoring developments and providing timely assessments in Afghanistan, this paper explores nowcasting quarterly GDP using machine learning (ML) with both traditional economic indicators and non-traditional data, such as satellite imagery of vegetation, precipitation, and $NO_2$ emission. It also addresses the lack of high frequency GDP data by leveraging the strong correlation between money supply and economic activity in Afghanistan's predominantly cash-based economy.

The findings provide important insights for Afghanistan. Results show that public investment, night-time light intensity, and agriculture-related indices (such as vegetation and rainfall) are strongly correlated with GDP growth and serve as effective predictors of economic activity.

The remainder of this paper is organized as follows: Section 2 reviews the literature on nowcasting economic activity, with a particular focus on low-income, fragile, and conflict-affected countries such as Afghanistan. Section 3 presents stylized facts about economic activities in Afghanistan. Section 4 describes the dataset. Section 5 describes the methodology. Section 6 discusses empirical results. Finally, Section 7 summarizes the key findings.

* The views expressed here are those of the authors and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

## 2. Literature Review

Nowcasting—forecasting the present—is an emerging field, with seminal contributions from Giannone et al. (2005). Its most prevalent application is estimating GDP, particularly in advanced economies, though efforts have been extended to nowcasting other economic indicators, such as inflation. Nowcasting leverages high frequency indicators—which are often available before official statistics are released—to provide timely insights into the trajectory of economic activity. Its effectiveness, however, hinges on the availability and quality of high frequency data.

Nowcasting economic activity in many developing economies presents significant challenges due to the lack of timely and high frequency macroeconomic indicators. The literature highlights the importance of non-traditional data sources, particularly in fragile and conflict-affected countries or those with limited statistical capacity. Advances in satellite data and machine learning (ML) offer promising opportunities for nowcasting GDP in data-scarce countries with improved accuracy (McSharry and Mawejje, 2024). Satellite imagery—particularly night-time light, nitrogen dioxide ($NO_2$), and vegetation indices—are increasingly used as proxies for economic activity due to their high correlation with GDP at global, regional, national, and even sub-national levels Forbes (2013); Ezran et al. (2023); Gibson et al. (2021);. While night-time light effectively captures economic activities, including the informal sector at night (though it is less accurate for agricultural activities), $NO_2$ measurements taken during the day reflect daytime activities. Additionally, studies have demonstrated robust correlations between vegetation indices and agricultural activities (Hu and Xia, 2019; Puttanapong et al., 2023; Bolivar, 2024).

Nowcasting GDP has demonstrated stronger performance than traditional forecasting methods. For instance, IMF (2022) provides monthly updates on economic developments for Sub-Saharan Africa. Tiffin (2016) applies machine learning techniques to nowcast Lebanon's GDP, addressing delays in data publication. Polo et al. (2025) uses machine learning to nowcast non-Oil GDP in GCC countries. Similarly, Akbal et al. (2023) propose a panel approach that utilizes statistical relationships from data-rich countries to estimate quarterly GDP for nations lacking such statistics, employing readily available indicators.

Several studies have leveraged high frequency data to estimate economic activity in Afghanistan given to the lack of timely official data. IMF (2019) employs nightlight data to estimate both sub-national and national GDP levels, finding that the estimated sub-national GDP closely reflects the distribution of major economic centers, with national estimates highly correlated with official economic data. Cabanillas et al. (2024) filter out nightlight data from military installations, revealing a 75 percent correlation between civilian nightlight data and economic activity. Saenger et al. (2024) assess the economic impact of the Taliban's takeover in August 2021 by comparing nightlight data from Afghanistan with that of neighboring countries. Galdo et al. (2020) examines the effect of conflict on Afghanistan's economy, focusing on formal, informal, and illicit

activities. Using high frequency data such as night-time light, mobile phone traffic, and poppy cultivation, their district-level analysis from 2012 to 2016 shows that heightened conflict intensity significantly reduces formal economic activity in the following quarter, while illicit activities increase after two quarters.

This paper contributes to the literature on economic monitoring in data-scarce environments by utilizing a nowcasting framework that combines machine learning and satellite imagery to estimate quarterly GDP for Afghanistan.

## 3. Stylized facts

Economic activity in Afghanistan has been significantly shaped by international aid and the presence of foreign troops. Repeated conflicts throughout its history have left deep economic scars, while foreign assistance has contributed to substantial fluctuations in economic activity. This volatility is further compounded by an underdeveloped agricultural sector that remains highly vulnerable to climate shocks and natural disasters.

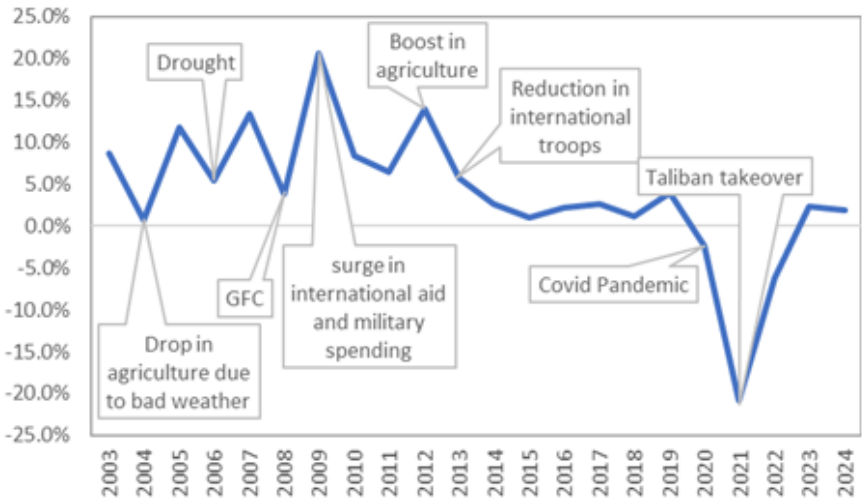
Afghanistan's Economic growth has exhibited significant volatility over the past two decades (Figure 1), reflecting exposure to a range of shocks, such as political events, the global financial crisis, the COVID-19 pandemic, droughts, and natural disasters. In 2004, the growth rate fell to below 1 percent, primarily due to adverse weather affecting agriculture. Major fluctuations are often linked to political developments that influenced the presence of foreign military forces and levels of international aid. For example, in 2009, the growth rate surpassed 20 percent, coinciding with a surge in international troop deployments and military spending, increased international aid, and a rebound in agricultural activity driven by favorable and well-distributed rainfall.[1] As the number of foreign troops and the volume of aid began to decline in 2012, growth rates moderated, averaging around 2.6 percent between 2012 and 2018.[2] Following the Taliban takeover in 2021, a substantial reduction in non-humanitarian international aid triggered a severe recession, with GDP contracting by about 21 percent in 2021 and a further 6.2 percent in 2022. However, the economy began to recover, experiencing modest growth in 2023 and 2024.

Over the past two decades, the service sector has emerged as the largest contributor to Afghanistan's economy, even though agriculture continues to employ the majority of the workforce (Figure 2). Aid-driven demand, particularly in sectors such as construction and transport, has shifted economic activity away from traditional agriculture toward services. However, the latter's share of GDP has declined in recent years due to a reduction in international grants, which is linked to the lack of recognition of the *de facto* authorities

[1] Source: DW: Afghan troop levels.
[2] See the IMF Article IV Consultation 2017.

Figure 1: Annual Economic Growth in Afghanistan, 2003-2024

(*Percent*)




Source: IMF, WEO database and IMF staff calculations. Data up to 2021 was based on calendar year after which they were based on the solar year.

as Afghanistan's government by the international community. In 2023, the service sector accounted for 47 percent of GDP, down from a peak of about 60 percent in 2016, with the sharpest drop occurring in 2021 (to 47 percent compared to 55 percent in 2020). Conversely, agriculture's share has risen since 2021, despite a prolonged drought from 2021 to 2023. Meanwhile, industry has remained modest and stable at around 14 percent of GDP.

On the demand side, private consumption remains the primary driver of GDP growth, although its impact is often partially offset by imports (Figure 3).[3] Changes in private consumption are partially attributed to foreign assistance: during periods of economic expansion, consumption rises, positively impacting GDP, but imports also increase, which dampens overall growth. This pattern reverses during recessions, when both consumption and imports contract.

[3] Data from 2022 are in Solar Year.

Figure 2: Sectoral Decomposition of Actual GDP, 2017-2024

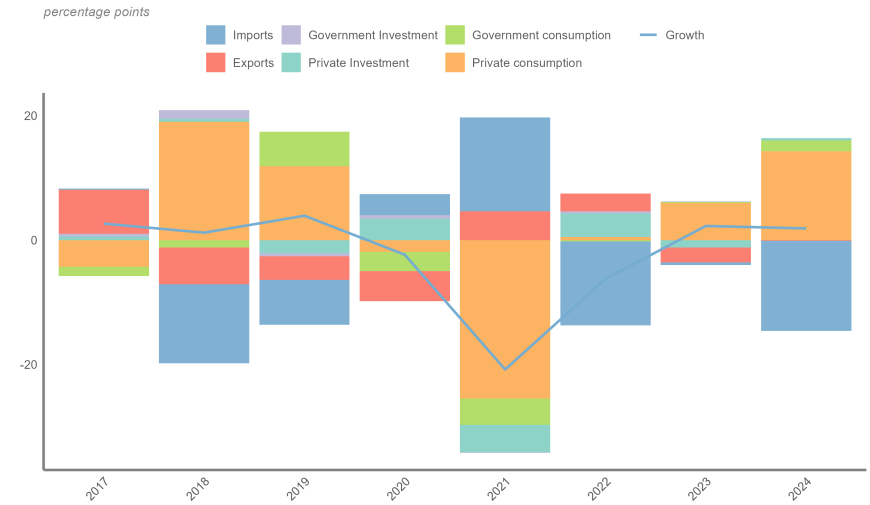
(*Sectoral Shares, Percent*)




Sources: IMF, WEO database and IMF staff calculations. Data from 2016 to 2021 followed calendar year (January-December), while data from 2022 onward followed Fiscal Year (April to March)

Figure 3: Contributions to Actual GDP — Demand Side, 2017-2024

(*Percent*)




Source: IMF, WEO database and IMF staff calculations. Data from 2016 to 2021 followed calendar year (January–December), while data from 2022 onward followed Fiscal Year (April to March).

## 4. Data

This paper utilizes a set of macroeconomic and non-traditional data (Table 1). This section presents the dataset including the construction of the quarterly GDP series.

Table 1: Macroeconomic Indicators and Non-Traditional Data

| **Macroeconomic Indicators** | **Non-traditional Data** |
|---|---|
| Goods balance | Nightlight (NTL) |
| Domestic Revenue | Nitrogen Dioxide ($NO_2$) |
| Exchange rate (REER) | NDVI (vegetation index) |
| Consumer Price Index | Rainfall |
| Public Investment | |
| Net foreign assets | |

### *4.1. Quarterly GDP Series*

Afghanistan's economic activity is highly dependent on cash transactions, a relationship that is directly reflected in the money supply (M2). Annual nominal GDP and M2 exhibit a strong correlation of about 98 percent between 2006 and 2024 and the correlation between M2 and real GDP is almost 88 percent for the same period [4] (Figure 4). The economy operates predominantly on a cash basis, with a substantial share of transactions conducted outside the banking channels. This reliance on cash underpins a direct link between money supply and economic activity, as measured by GDP through consumer spending and business investment. Moreover, a significant portion of Afghanistan's informal economy further limits the coverage of official economic indicators.

To construct a quarterly GDP series, this paper interpolates quarterly data from the annual figures by leveraging quarterly movements in $M2$ (deflated by CPI), which serves as a reliable proxy for economic activity. The conversion of the GDP series from annual to quarterly frequency was carried out using the Chow-Lin method,[5] resulting in interpolated quarterly real GDP data that closely track the fluctuations of actual annual real GDP from 2007 to 2024 (Figure 4). The interpolated GDP data will be used to nowcast GDP growth later in the paper.

### *4.2. Other Macroeconomic Data*

A range of high frequency macroeconomic indicators for Afghanistan—released monthly and quarterly, typically with an average delay of three months—show strong correlations with both nominal and real GDP. [6] As illustrated in Figure 5, domestic fiscal revenues, public investment, and the Consumer Price Index (CPI) display strong positive correlations with real GDP. In contrast, the merchandise balance exhibits the most negative correlation over the sample period, reflecting Afghanistan's heavy reliance on imports.[7]

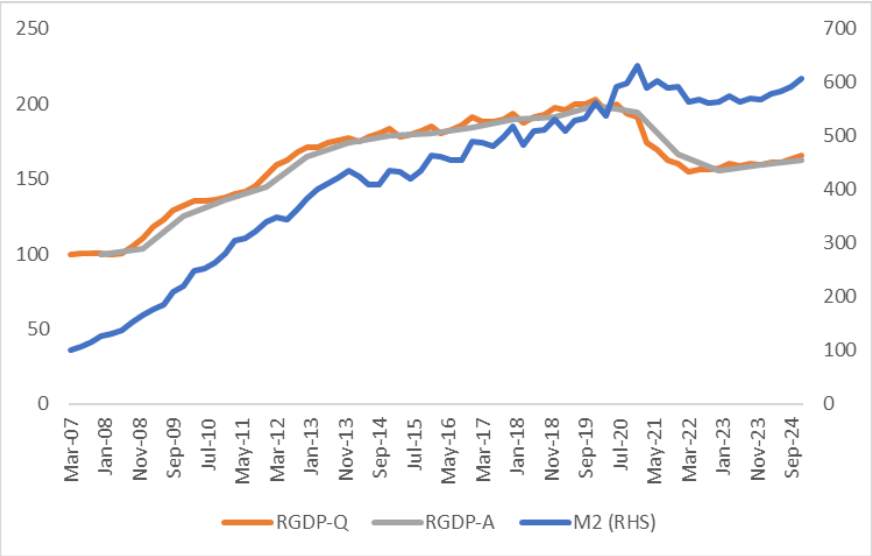
[4] The correlation between M2 and nominal GDP growth rates is also high around 79 percent.

[5] The Chow-Lin method is a regression-based statistical technique used to interpolate low-frequency time series data (e.g., annual figures) to higher frequencies (e.g., quarterly or monthly).

[6] While non-traditional indicators, such as prices data, may be utilized to generate monthly nowcast, we opt to concentrate on quarterly nowcasting to leverage the complete set of available macroeconomic indicators.

[7] Net foreign assets (NFA) are also negatively correlated with economic growth; however, NFA data have become unreliable after 2021 due to the bulk of foreign reserves are frozen, potentially introducing distortions into this relationship. However, we kept NFA

Figure 4: Afghanistan's Actual Annual Real GDP Vs Quarterly Interpolated GDP, 2007-2024




Source: IMF, WEO database and IMF staff calculations.

### 4.3. *Non-Traditional (satellite) Data*

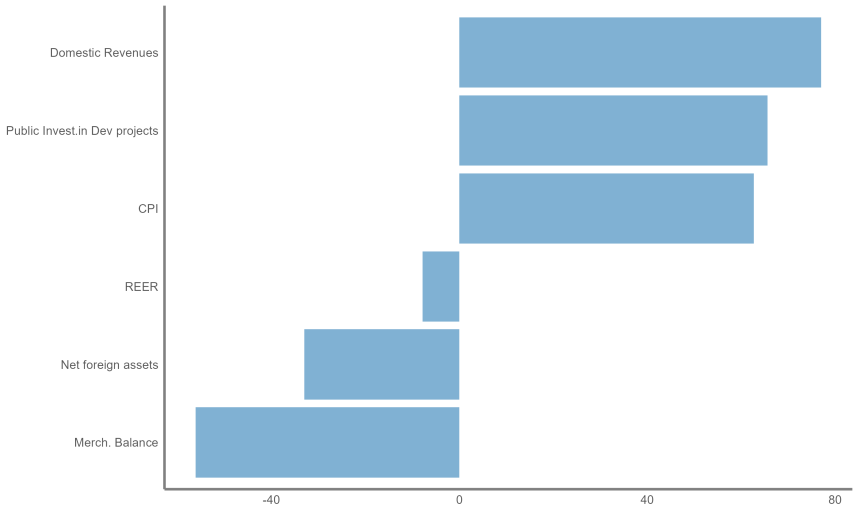
***Night-time Light (NTL)*** is widely used as a proxy for economic activity, although it does not fully capture output in certain sectors, such as agriculture. Data from NASA's Suomi National Polar-orbiting Partnership (SNPP) satellite, specifically the Visible Infrared Imaging Radiometer Suite (VIIRS), provide daily, high-quality NTL measurements after correcting for moonlight and atmospheric distortions. Figure 6 displays a map of the mean NTL intensity across Afghanistan (panel a), and changes between 2012 and 2024 (panel b), which reflect percentage variations in intensity. In panel (b), blue areas indicate decreases in NTL, red areas show increases, and white areas denote little or no change—primarily corresponding to regions with minimal human settlement or economic activity. Overall, NTL data show a moderate increase between 2012 and 2024, particularly in regions near Kabul.

***Nitrogen Dioxide ($NO_2$)*** is a key trace gas generated by both fossil fuel combustion (from coal, oil, and gas) and natural processes such as wildfires and lightning. As $NO_2$ tends to accumulate near its source and has an atmospheric lifespan of less than a day, its concentration is a reliable indicator of localized combustion activity and emissions (Seinfeld and Pandis, 2016). The $NO_2$ data are derived from vertical column densities (VCDs) measured by the Ozone Monitoring Instrument (OMI) aboard NASA's Aura satellite Lamsal et al. (2008, 2010). Figure 7 shows both the mean $NO_2$ emission values and changes in their spatial distribution

in the model as it captures important external and foreign exchange dynamics.

Figure 5: Correlations of Macro Variables with Actual GDP

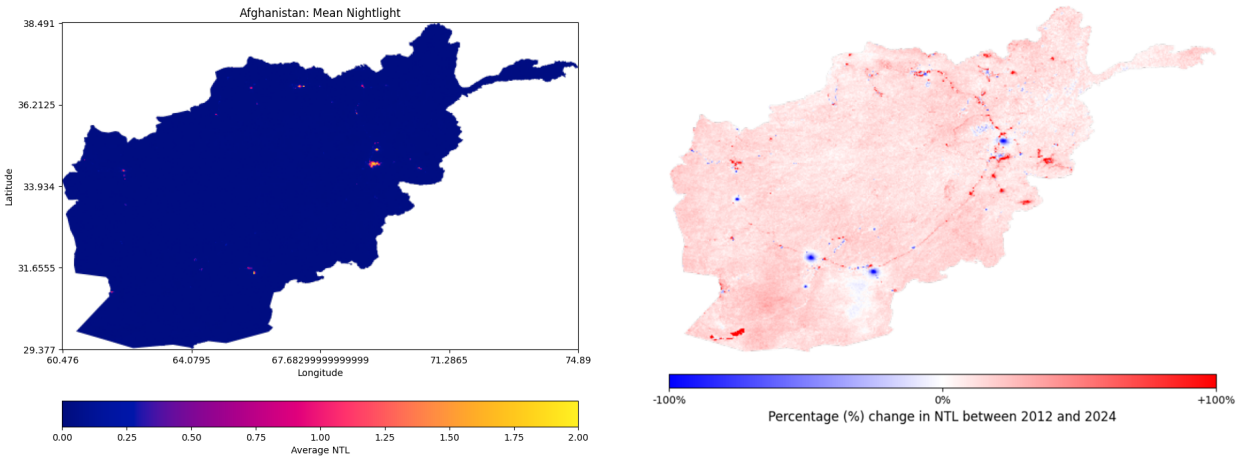
(*Percent*)




Source: IMF, WEO database and IMF staff calculations.

Figure 6: Afghanistan Nightlight

(a) (*mean*) (b) (*changes: 2012—2024*)

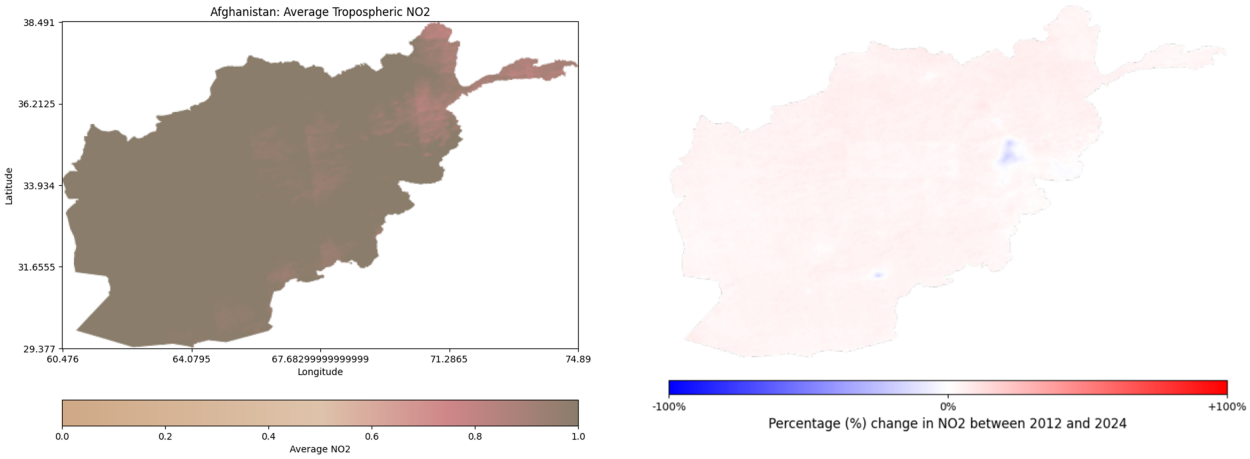



Source: IMF, WEO database and IMF staff calculations.

across Afghanistan between 2012 and 2024. Blue areas represent decreases in $NO_2$ emissions, red areas indicate increases, and white areas denote regions with little or no change, typically corresponding to sparsely populated areas or economic activity. Overall, the data reveal negligible changes in $NO_2$ emissions from 2012 to 2024, consistent with Afghanistan's limited and underdeveloped manufacturing sector.

Figure 7: Afghanistan *NO*2 Emissions

(a) (*mean*) (b) (*changes: 2012—2024*)




Source: IMF, WEO database and IMF staff calculations.

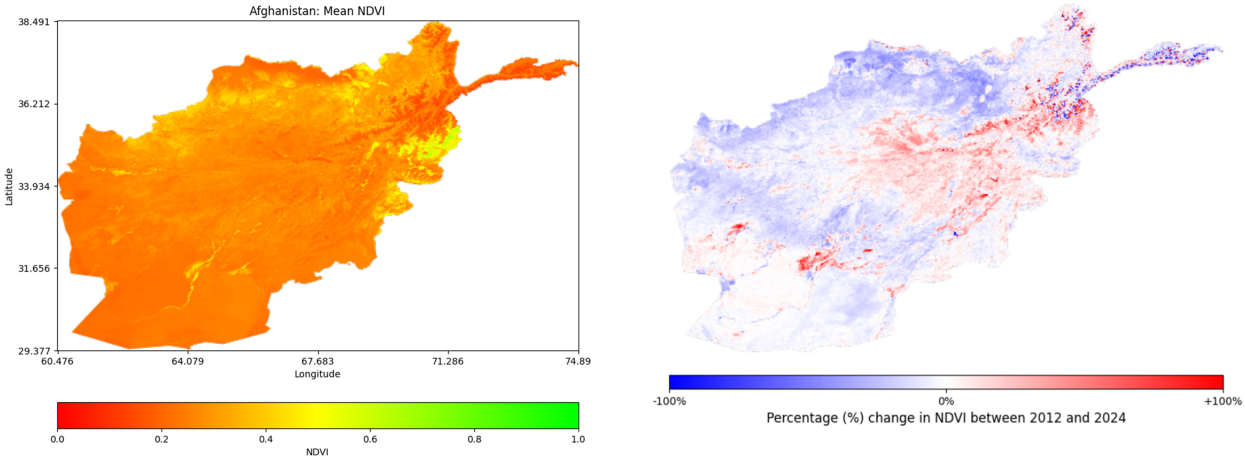
***Agriculture Indices:*** Agriculture plays a vital role in Afghanistan's economy, accounting for approximately one third of GDP and serving as a key source of income for much of the population. However, the sector remains highly susceptible to climate shocks, as evidenced by Afghanistan's recurrent episodes of severe drought, including the one from 2021 to 2023. To assess agricultural performance, this paper utilizes two key indicators: a vegetation index and rainfall data, both are closely linked to agricultural output.

- Figure 8 presents the Normalized Difference Vegetation Index (NDVI), a widely used metric for monitoring agricultural activity and assessing vegetation health and density. It highlights the impact of successive droughts, showing reduced agricultural activities compared to 2012.
- Figure 9 illustrates the rainfall index, which serves as a proxy not only for agricultural production but also for broader climate-related shocks affecting the economy.

Overall, Figure 9 depicts the temporal trends of non-traditional data in Afghanistan. A particularly sharp decline in NTL is evident in 2020, which is attributed to the impact of COVID-19. These observable changes in NTL, which show a strong correlation with economic activity, highlight its value as a reliable predictor of economic conditions. The figure also tracks the evolution of $NO_2$ growth over time, revealing notable fluctuations in emissions that are closely associated with shifts in economic activity.

Figure 8: Afghanistan *NDVI*

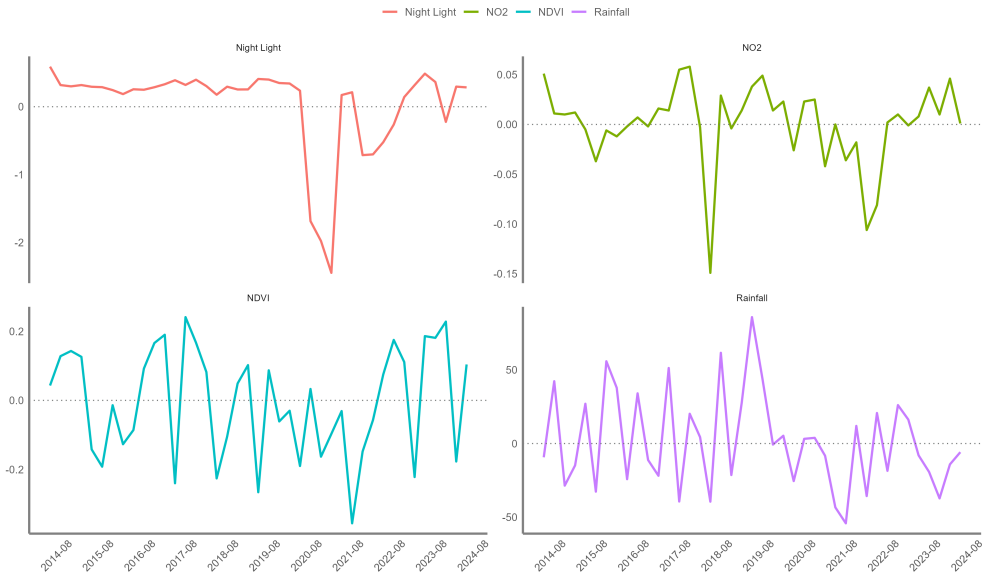
(a) (*mean*) (b) (*changes: 2012—2024*)




Source: IMF, WEO database and IMF staff calculations.

Figure 9: Afghanistan Non-Traditional Data Growth Over Time

(*percentage change, yoy*)




Source: IMF, WEO database and IMF staff calculations.

## 5. Methodology

This paper adopts the methodology outlined in Barhoumi et al. (2022), evaluating a range of machine learning models based on out-of-sample predictive performance. Specifically, the first 85 percent of the data is used as the training set, while the remaining 15 percent serves as a holdout set for model evaluation. This "horserace" approach facilitates the selection of the best-performing model. Each model undergoes hyperparameter tuning prior to evaluation on the holdout set. Hyperparameters, which influence model complexity and accuracy, are optimized using three-fold cross-validation within the training set. In this procedure, the training data is randomly divided into three folds: models are trained on two folds and validated on the third, resulting in three error estimates. The Root Mean Squared Error (RMSE) serves as the performance metric guiding hyperparameter selection. After the optimal hyperparameters are identified, the final model is trained on the full training set and used to predict GDP growth in the holdout set.[8]

The models evaluated in this study include Random Forest, Bagged Trees, XGBoost, Cubist Regression, Support Vector Machines (SVM) with both Linear and Polynomial Kernels, Decision Tree, and Ordinary Least Squares (OLS) regression. This selection spans a diverse range of machine learning approaches frequently applied in predictive analytics. Ensemble methods such as Random Forest and Bagged Trees enhance prediction accuracy by aggregating the outputs of multiple decision trees trained on bootstrapped samples, thereby reducing variance. XGBoost, a robust gradient boosting technique, incrementally optimizes residual errors to improve predictive performance. Cubist Regression blends rule-based models with boosted regression trees to achieve high numerical accuracy. SVMs, employing either linear or polynomial kernels, construct hyperplanes to distinguish classes or predict continuous outcomes in high-dimensional spaces. Decision Tree models offer interpretable, rule-based predictions but may be susceptible to overfitting. OLS serves as a benchmark linear regression model, quantifying direct relationships between predictors and the response variable.

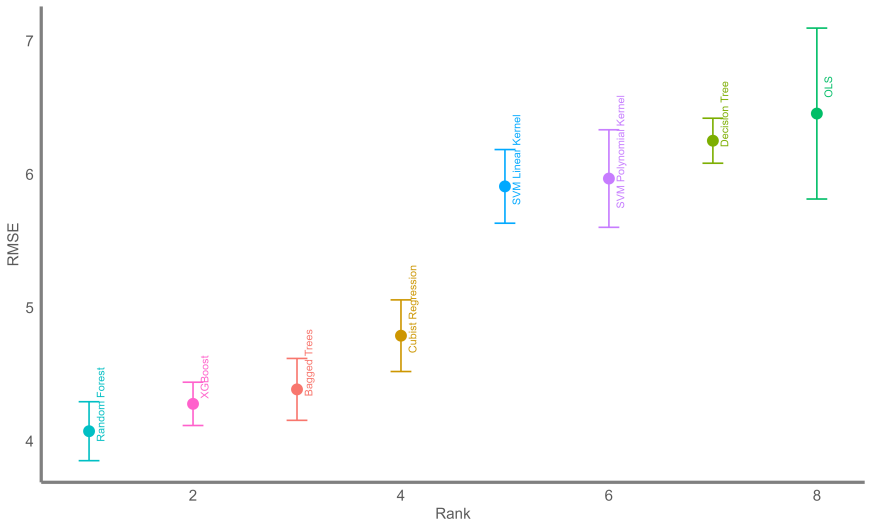
Among these machine learning models, the Random Forest (RF) algorithm achieved the lowest RMSE (Figure 10), indicating superior predictive accuracy. As outlined in Breiman (2001), the RF algorithm constructs an ensemble of randomized decision trees and averages their predictions (Figure 11). This methodology is particularly well-suited for multivariate regression tasks, offering notable advantages in terms of simplicity and predictive power across datasets of varying size and dimensionality. The RF model effectively captures nonlinear relationships and complex interactions among variables, mitigates overfitting by utilizing random subsets of data and features for each tree, and demonstrates robustness to outliers and

[8] RMSE measures the distance between actual time series and predicted values, making it a common metric for evaluating prediction accuracy. RMSE is employed for both hyperparameter tuning in the training sample and for model selection in the holdout set horserace.

missing values.

Figure 10: Comparing the Performance of Various Machine Learning Models

(*horserace*)




Source: IMF, WEO database and IMF staff calculations.

In a dataset with $n$ samples, $x$ explanatory variables (traditional and non-traditional data), and $y$ as the target variable (quarterly GDP), the RF algorithm operates as follows:

1. Initialize: The number of decision trees, $T$; the number of explanatory variables to be randomly selected at each node, $x'$; and the maximum depth of each tree, $d$.
2. For $t = 1$ to $T$: Randomly select $x'$ from the original $x$; and construct a decision tree using the selected explanatory variables with a maximum depth of $d$.
3. For a randomly selected $x'$, predict the target variable $y'$ as:

$$y' = \frac{1}{T} \sum_{t=1}^{T} f_t(x') \tag{1}$$

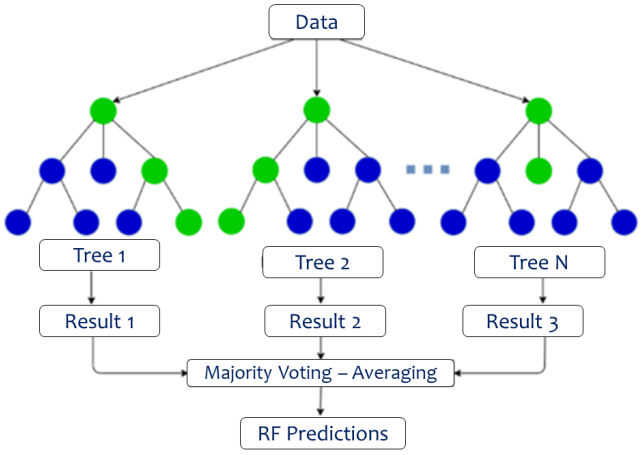
where $f_t(x')$ is the prediction of the $t^{th}$ decision tree.

## 6. Empirical Results

The alignment between the actual and nowcasted GDP growth (y/y) is strongly indicative of the model's reliability, as illustrated in Figure 12. For most of the sample period, the nowcasted GDP closely tracked the actual GDP data. The most notable divergence occurred in 2021 (Taliban's takeover), when the nowcasted GDP indicated a smaller contraction than what was reflected in the official GDP figures. This discrepancy may be attributable to economic activities that are not captured by official statistics but are detected through

Figure 11: Random Forest Prediction Process

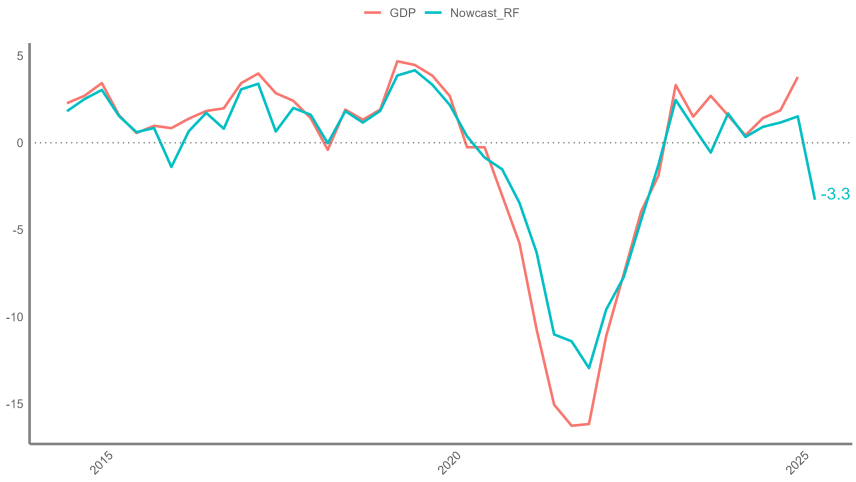
(*illustration*)




Source: IMF, WEO database and IMF staff calculations.

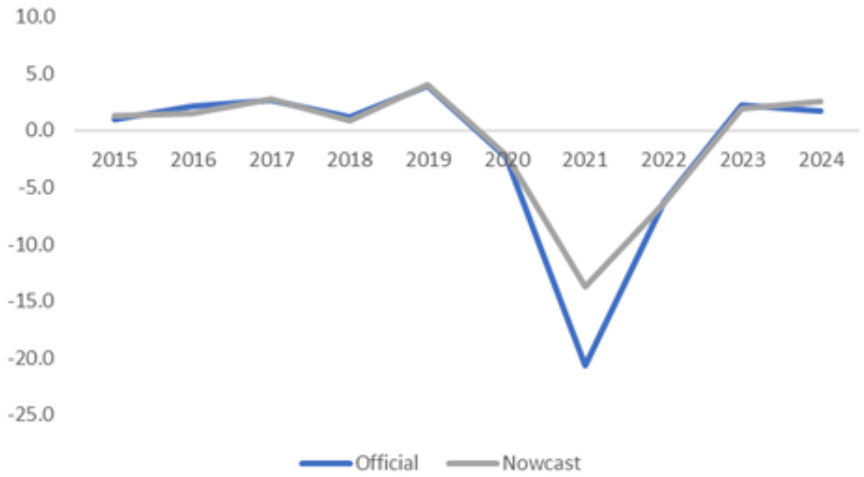
non-traditional data sources. These co-movements, as well as the 2021 deviation, become even more apparent when quarterly nowcasts are aggregated to annual levels (Figure 13). The nowcasting results for 2025Q1 indicate a year-on-year growth rate of -3.3 percent. This significant decline in 2025Q1 is likely attributed to the reduction in international aid, which plays a critical role in Afghanistan's economy. Although much of this aid does not flow directly through the government budget, the Afghan economy remains highly dependent on it. As economic activity slows, the government's capacity to mobilize revenue and finance investment and other expenditure is further diminished.

Figure 12: Afghanistan Quarterly Real GDP Growth: Actual versus Nowcast




Source: IMF, WEO database and IMF staff calculations.

Figure 13: Afghanistan Annual GDP Growth: Actual versus Nowcast

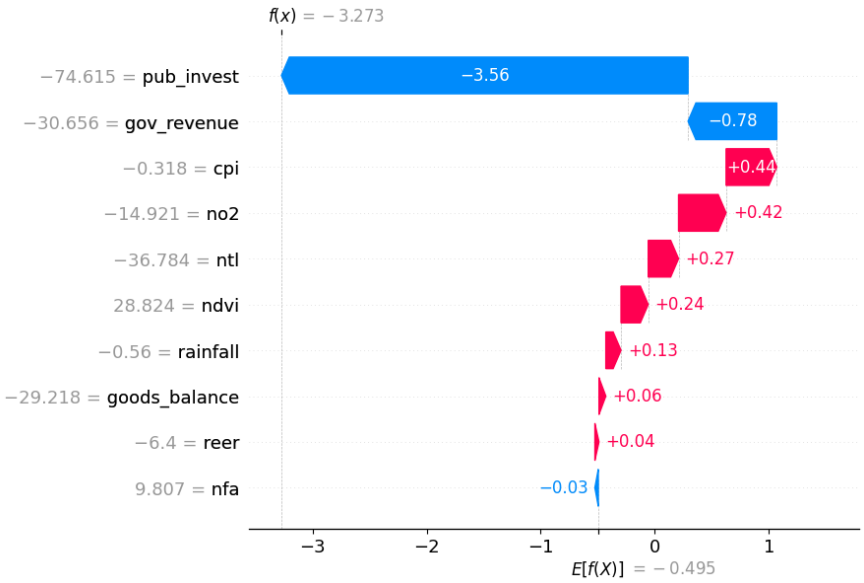



Source: IMF, WEO database and IMF staff calculations.

The SHAP bar plot in Figure 14 demonstrates the contribution of each feature to the 2025Q1 nowcast value. Features are ranked by their influence on the prediction, with the direction and magnitude of each bar indicating whether the feature increases or decreases the predicted value. In this analysis, Public investment stands out as a significant contributor, exhibiting a wide range of positive SHAP values and highlighting its role as a key driver in the model. Additionally, Figure 15 presents the historical SHAP decomposition, illustrating the annual contributions of each feature to the GDP nowcast over time. Notably, public investment, night-time light intensity, and agriculture-related indices consistently rank among the most influential predictors.

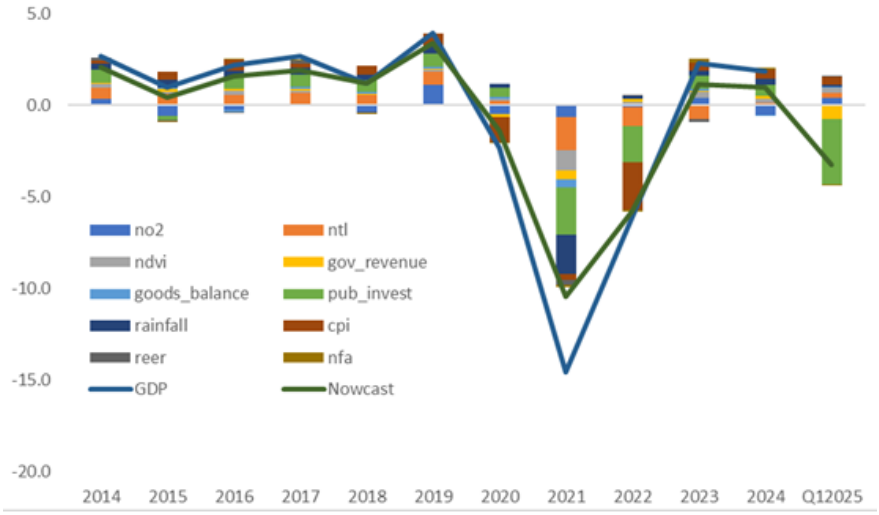
Historically, public investments—particularly in development projects—have been heavily reliant on foreign aid. Prior to 2022, on-budget grants averaged nearly 7 percent of GDP from 2010 to 2020 and accounted for close to 55 percent of total government revenues over that period. This substantial inflow of aid enabled Afghanistan to finance public investments, which have served as the primary engine of economic growth in the context of a nascent private sector. Consequently, increased public investment in development projects has consistently contributed positively to overall economic activity and GDP growth. While correlation measures the strength of the association between a variable and GDP, SHAP values measure the variable's incremental contribution to the model's prediction after accounting for the information contained in all other predictors. As a result, a variable such as government revenues may be highly correlated with GDP yet have a small SHAP contribution if its predictive content is largely captured by other explanatory variables.

Figure 14: Contributions to 2025Q1 GDP Nowcast




Source: IMF, WEO database and IMF staff calculations.

Figure 15: Contributions to GDP Nowcast by year




Source: IMF, WEO database and IMF staff calculations.

Overall, the analysis demonstrates that combining traditional macroeconomic indicators with non-traditional indicators can predict both the magnitude and direction of GDP growth in Afghanistan with reasonable accuracy. Across many quarters, night-time light intensity and public investment emerged as the most influential drivers of the model's predictive ability, underscoring their central role as indicators of economic growth. However, more recently, public investment appears to have assumed a relatively more prominent role. Notably, negative growth in either of these indicators tends to be associated with negative GDP growth.

Several caveats are worth noting. The interpolated variable used in this study is only a proxy for aggregate economic activity and may not fully reflect developments in Afghanistan's large informal economy. In addition, the short sample period constrains estimation and may limit the robustness of the results. Data quality issues affecting NFA statistics may also introduce measurement error into the analysis. Accordingly, the findings should be interpreted with caution and reassessed as longer and higher-quality data series become available.

## 7. Conclusion

Afghanistan faces a significant challenge in the absence of high frequency and timely GDP data, with annual figures often released after considerable delays. This constraint hampers effective economic surveillance and policymaking—a common issue prevalent among small, fragile, and conflict-affected states with limited statistical capacity. To address this gap, the paper utilizes an innovative methodology for estimating quarterly economic activity under such conditions, thereby enabling more timely economic assessments for Afghanistan. The approach interpolates quarterly GDP using money supply ($M2$), leveraging its strong correlation with economic activity in Afghanistan's predominantly cash-based economy. By integrating machine learning techniques with both traditional macroeconomic indicators and non-traditional data sources—such as satellite imagery—this paper provides insights into economic trends, offering a valuable tool for capturing the underlying direction of Afghanistan's economy at a relatively high frequency, even in the absence of comprehensive data.

The results indicate that public investment, night-time lights, and agriculture-related indices (e.g., vegetation and rainfall) are closely associated with GDP growth and serve as strong predictors of economic activity. In contrast, other indicators played a more limited role, reflecting the unique structure of Afghanistan's economy—marked by its dependence on agriculture and external aid.

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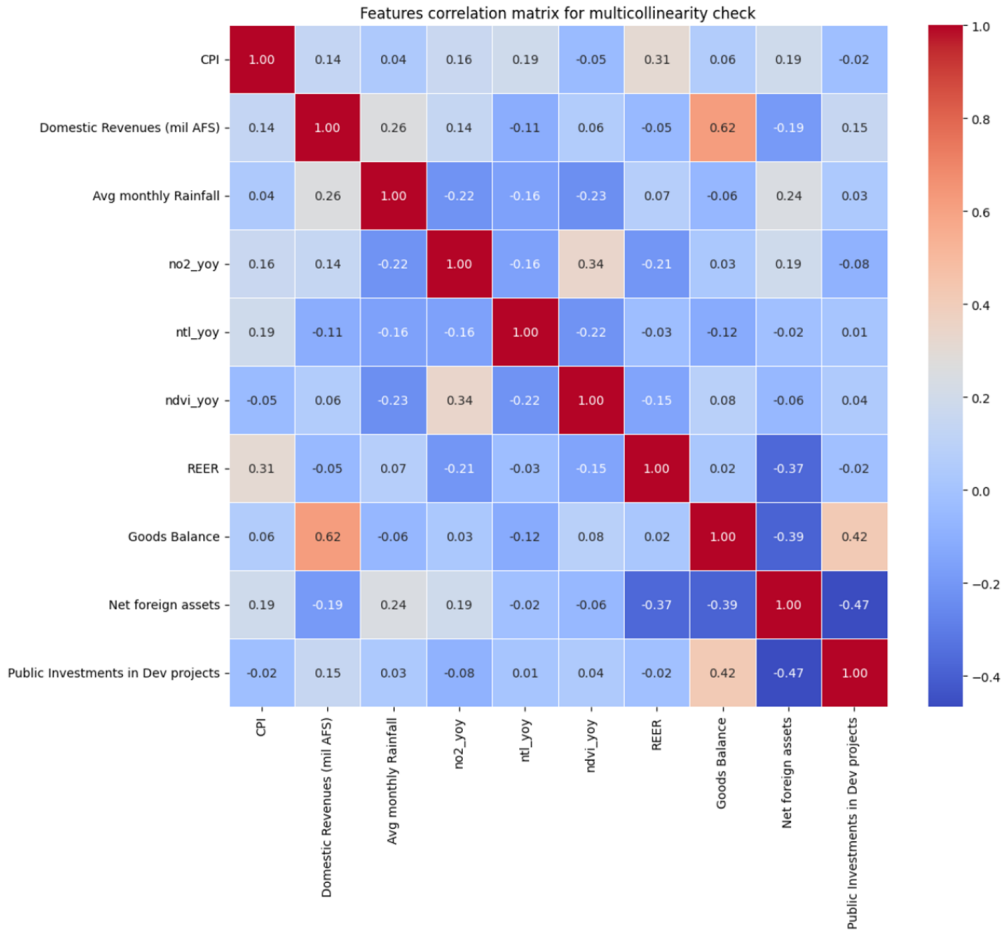
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## Appendix A.

### *Appendix A.1. Multicollinearity:*

Multicollinearity among predictors is less critical in machine learning than in traditional regression models. While regression focuses on model fit and can suffer from inflated standard errors due to multicollinearity, machine learning prioritizes predictive power. Thus, even if multicollinearity is present, it is unlikely to significantly affect our model predictions, as random forests are designed to minimize prediction error. Additionally, random forests split data based on individual features, which allows them to handle collinearity effectively. However, while the predictive power generally remains robust, collinearity may still influence the relative importance of predictors, potentially affecting model interpretability. Our analysis of the correlation between predictor pairs and indicates that multicollinearity is unlikely to impair the model's predictive performance materially (Figure A.1).

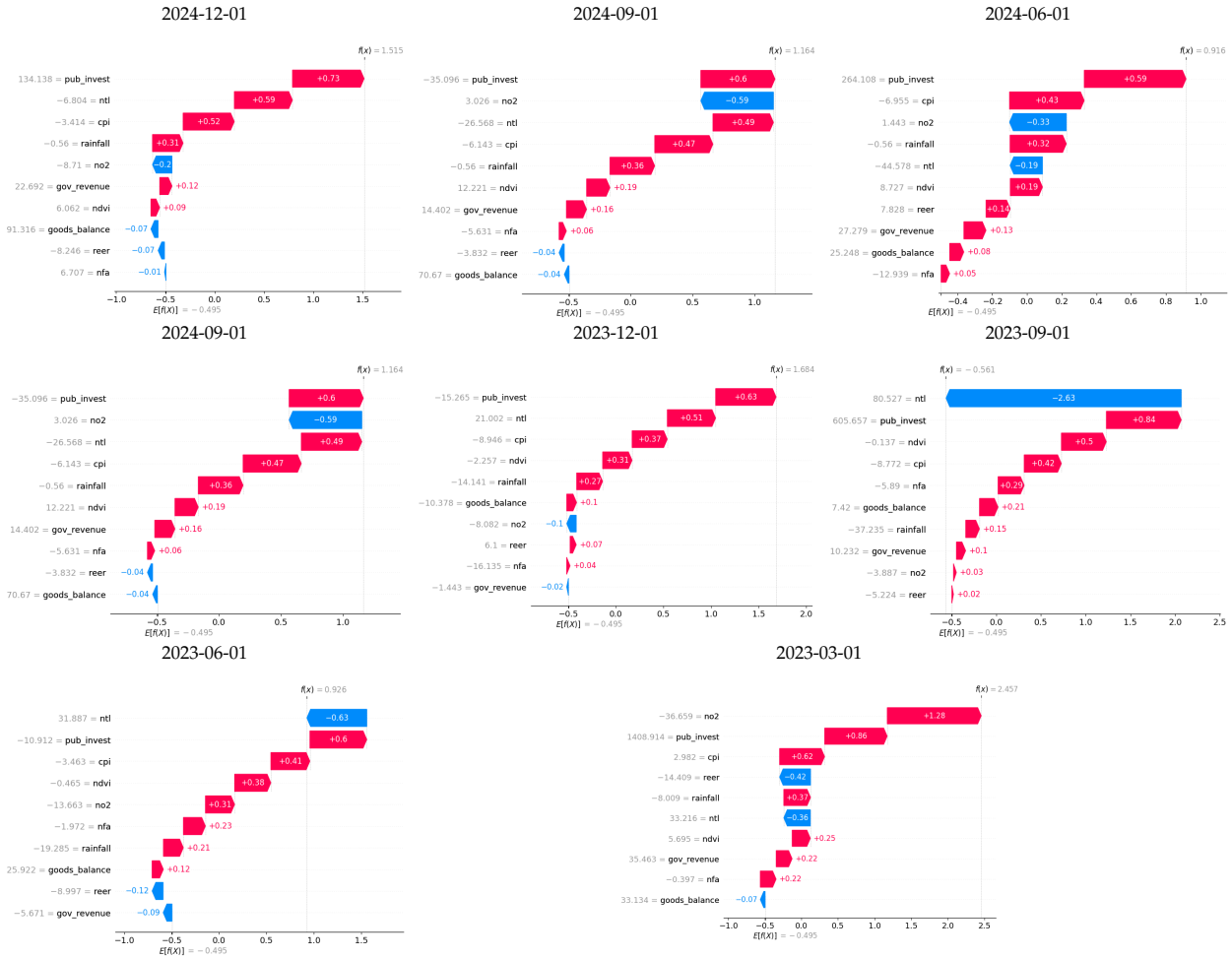
Figure A.1: Correlation Matrix




Source: IMF, WEO database and IMF staff calculations.

Figure A.2: Shapley Decomposition – Feature Contributions, 2023–2024

(*Feature Contribution Only*)




Source: IMF, WEO database and IMF staff calculations.



PUBLICATIONS

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